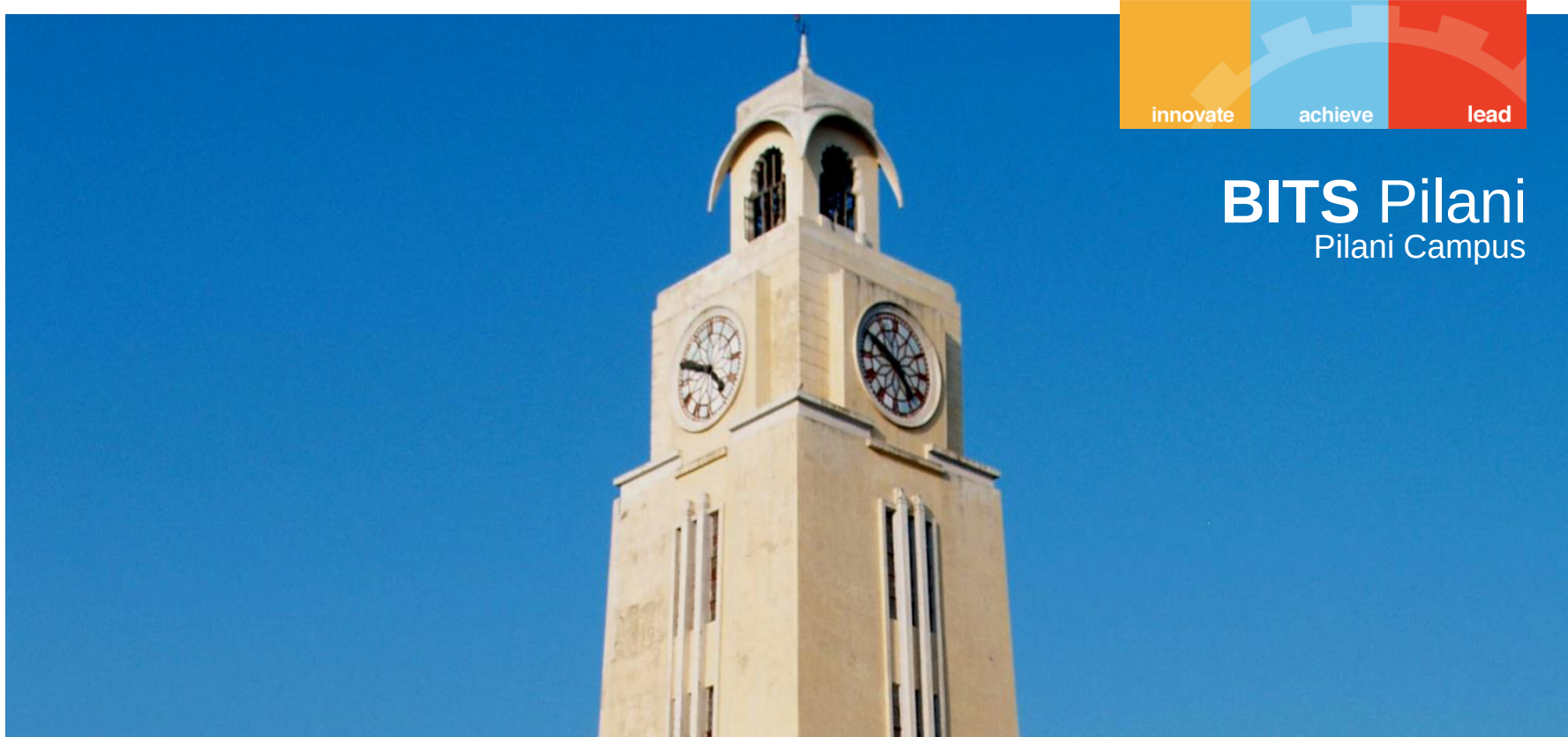




BITS Pilani
Pilani Campus



CHEM F111 : General Chemistry

Semester II: AY 2023-24

Lecture- 14, 16th February 2024, Friday

Last Class:

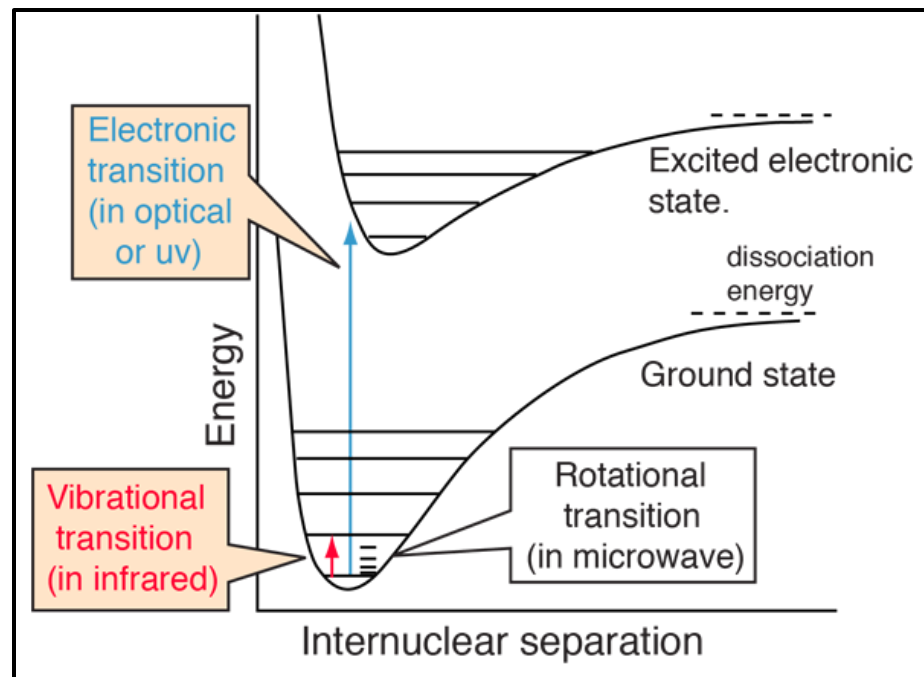
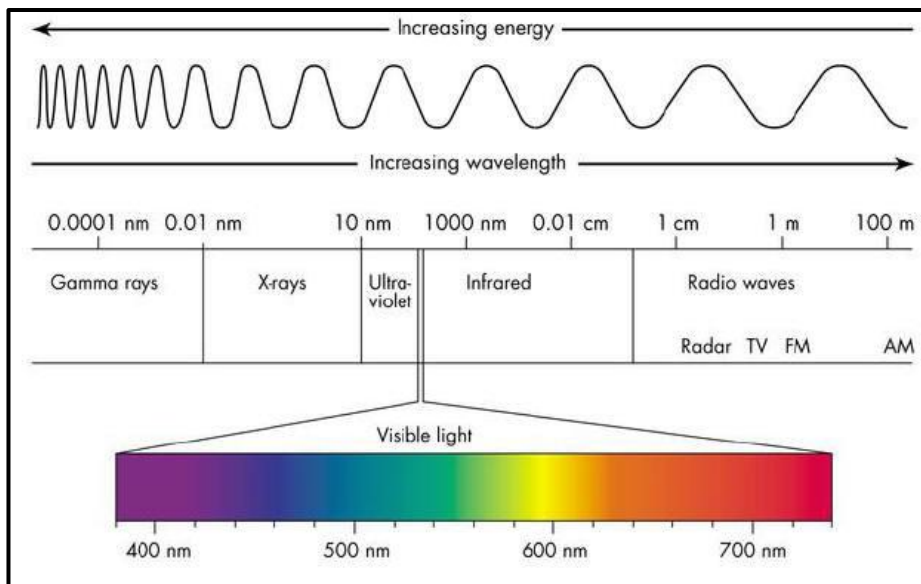
- Definition of spectroscopy; Introduction
- Electromagnetic spectrum and electromagnetic radiation;
- Interaction of electromagnetic radiation and molecule;
- Energy states of a molecule and transitions;
- Different types of spectroscopy;
- Electronic spectroscopy – UV-vis spectroscopy and emission spectroscopy; Jablonski diagram, absorption, fluorescence, phosphorescence; Radiative and non-radiative processes; timescales of various photophysical processes; examples of absorption and emission spectra of certain molecules

Recap: Molecular spectroscopy introduction

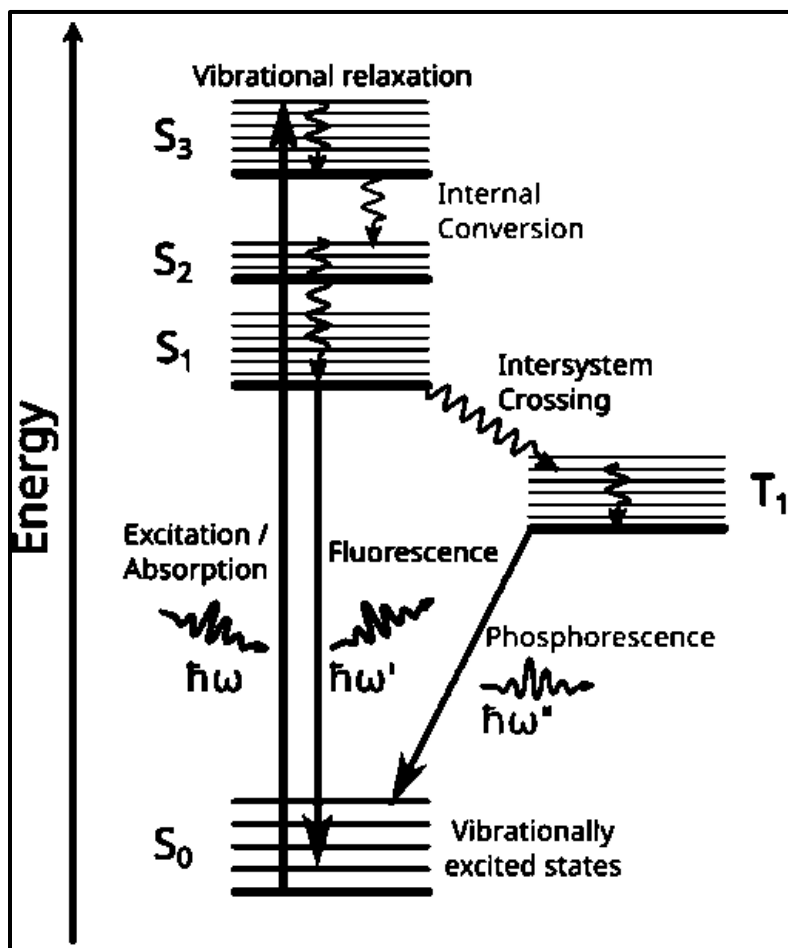
innovate

achieve

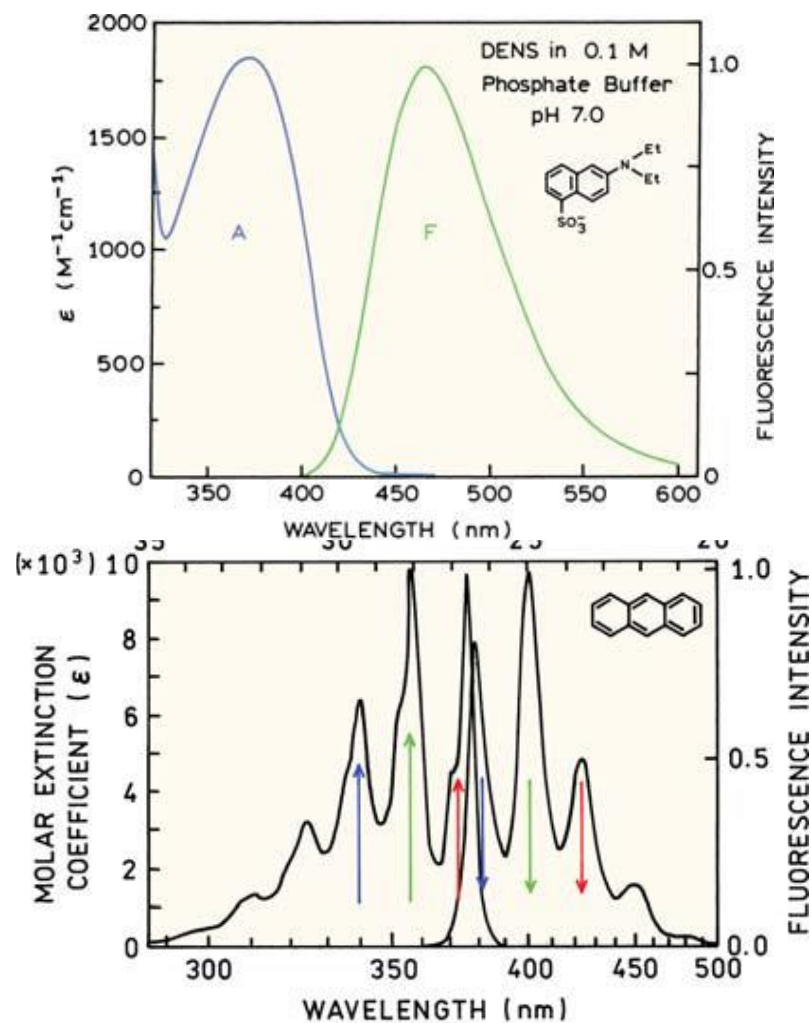
lead



Jablonski diagram



Spectra: Examples

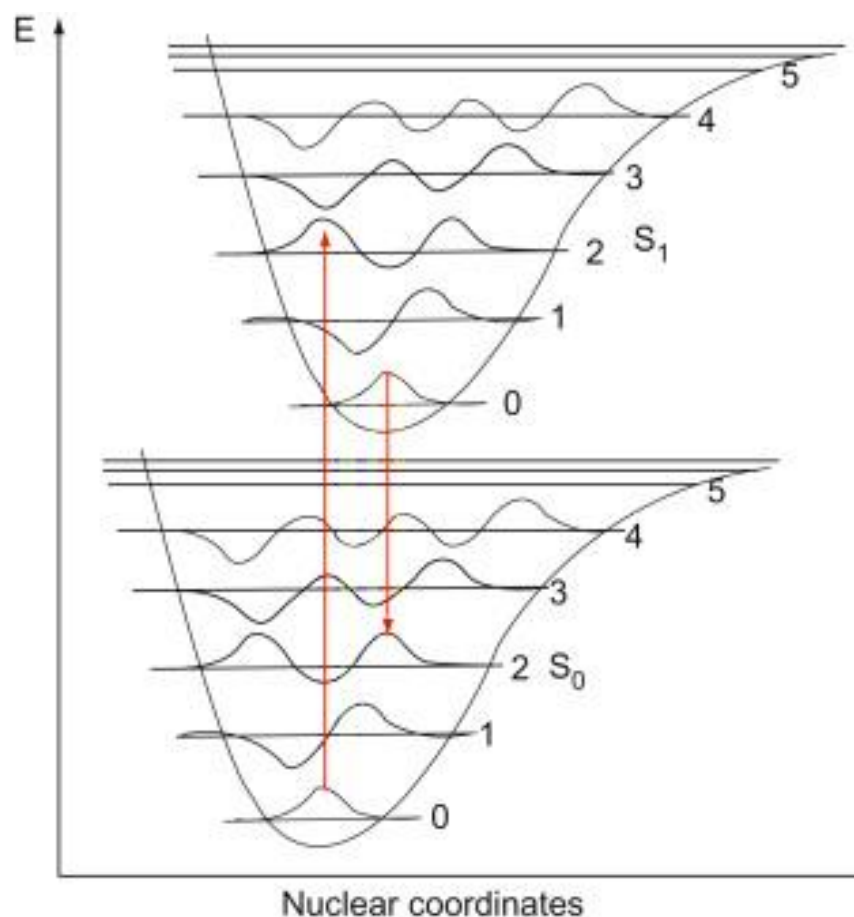


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***Electronic Spectroscopy:
Absorption (UV-Visible spectroscopy) and
Emission (Emission spectroscopy)***

Franck-Condon Principle

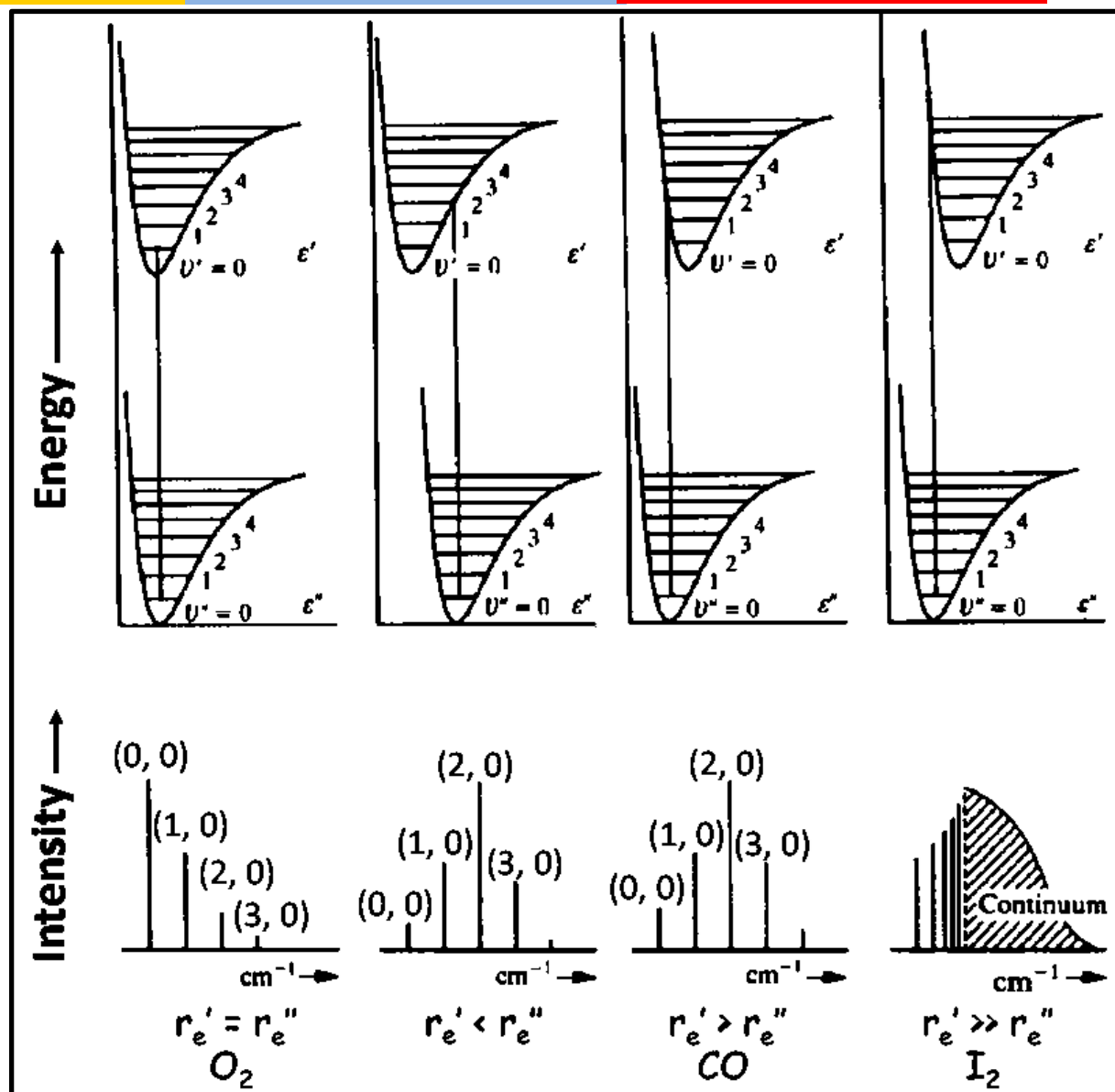
The **Franck-Condon principle** states that an electronic transition occurs more likely when the vibrational wave functions of the two levels overlap more significantly.



Electronic transitions are essentially instantaneous, so there is no time for appreciable motion of the nuclei. So, the transitions appear as vertical lines with no change in instantaneous internuclear distance during the electronic transition.

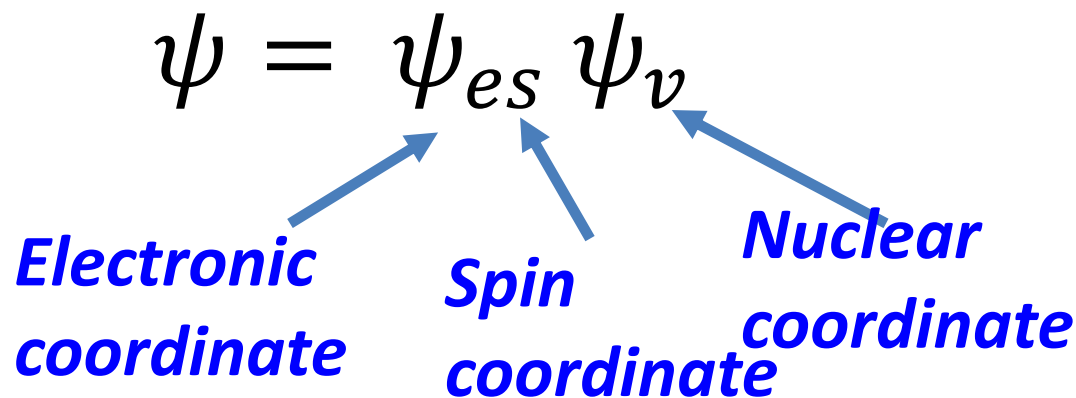
As a result of **Franck Condon principle**, some transitions are more probable for both absorption and emission than others.

Franck-Condon Principle - shapes of bands and intensities



Transition moment integral: $\int \psi'^* \hat{\mu} \psi d\tau$

Dipole moment operator, $\hat{\mu}$, can be divided into two components. One depends on nuclear coordinates ($\hat{\mu}_n$) and the other depends on electron coordinates ($\hat{\mu}_e$).

$$\psi = \psi_{es} \psi_v$$


Electronic coordinate

Spin coordinate

Nuclear coordinate

Transition moments

Transition moment integral: $\int \psi'^* \hat{\mu} \psi d\tau$

$$\psi = \psi_{es} \psi_v$$

$$\Rightarrow \int \psi_{e's'}^* \psi_{v'}^* (\hat{\mu}_n + \hat{\mu}_e) \psi_{es} \psi_v d\tau$$

$$\Rightarrow \int \psi_{e's'}^* \psi_{v'}^* \hat{\mu}_n \psi_{es} \psi_v d\tau + \int \psi_{e's'}^* \psi_{v'}^* \hat{\mu}_e \psi_{es} \psi_v d\tau$$

$\Rightarrow$

$$\int \psi_{e's'}^* \psi_{es} d\tau_{es} \int \psi_{v'}^* \hat{\mu}_n \psi_v d\tau_n + \underbrace{\int \psi_{v'}^* \psi_v d\tau_n}_{\text{FCF}} \int \psi_{e's'}^* \hat{\mu}_e \psi_{es} d\tau_{es}$$



0

Transition moment integral:

$$\int \psi_{v'}^* \psi_v d\tau_n \int \psi_e^* \hat{\mu}_e \psi_e d\tau_e \int \psi_{s'}^* \psi_s d\tau_s$$



*Franck-Condon
Factor*



*Orbital
selection rule*



*Spin
selection rule*

Selection rules for molecular transitions

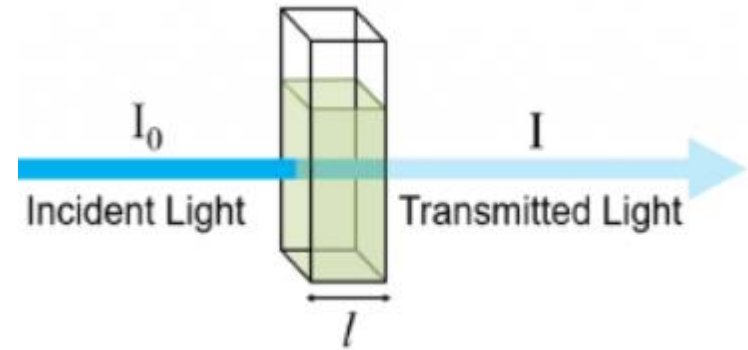


For molecules, the rules governing the transitions between two given energy states are:

- (i) $\Delta\Lambda = 0, \pm 1$; allowed changes in the component of the total angular momentum in the direction of the molecular axis.
- (ii) $\Delta S = 0$; the spin conservation rule.
- (iii) Symmetry properties of the energy states must be conserved.

- *Specific absorptions can be identified if electron is localized on a small group of atoms (Chromophores) E.g. C=O at 290 nm. ($n \rightarrow \pi^*$)*
- *Similarly, there is $\pi \rightarrow \pi^*$ transition in C=C systems.*
- *Conjugation increases absorption wavelength.*
- ***d-metal complexes***
 - *d-d transitions (weak) Mn^{2+}*
 - *Charge transfer transitions (strong–intense absorptions) MnO_4^-*

[This will be discussed in the next part of this course]



Beer-Lambert Law:

$$\text{Absorbance } \mathbf{A} = \log \frac{I_0}{I} = \epsilon [C] l;$$

$l = \text{length}; c = \text{concentration};$
 $\epsilon = \text{molar extinction coefficient}$
(unit: $1/(\text{conc.} \times \text{length})$)

$$\text{Transmittance } \mathbf{T} = \frac{I}{I_0} ; \quad \mathbf{A} = -\log T$$

Mixture of two species

Total absorbance at a given wavelength,

$$A = A_A + A_B$$

or

$$A = \epsilon_A [A]l + \epsilon_B [B]l$$

Total absorbance at λ_1 , $A_1 = \epsilon_{A1} [A]L + \epsilon_{B1} [B]L$

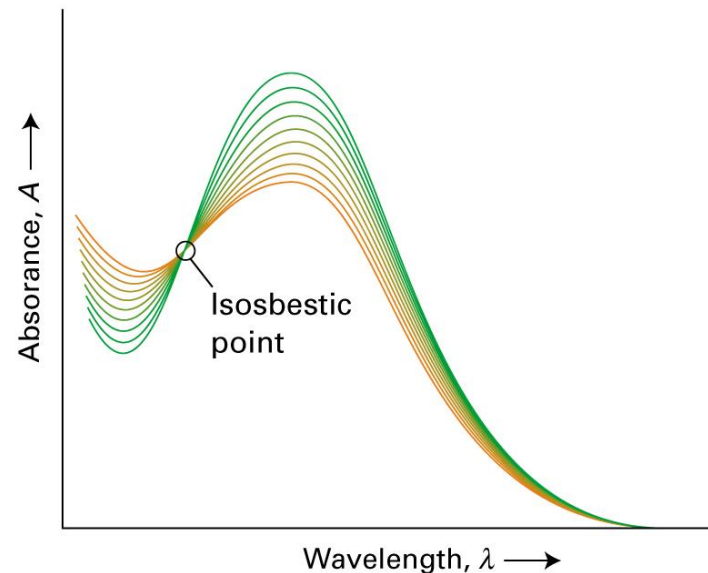
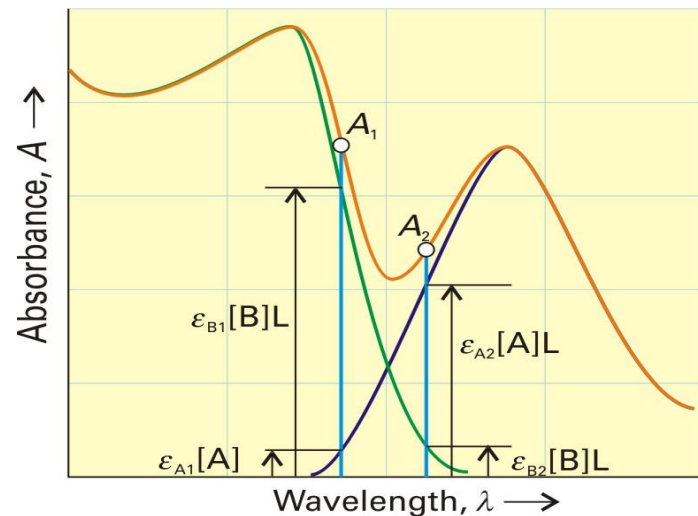
Total absorbance at λ_2 , $A_2 = \epsilon_{A2} [A]L + \epsilon_{B2} [B]L$

Solving these two equations, find individual concentration of A and B

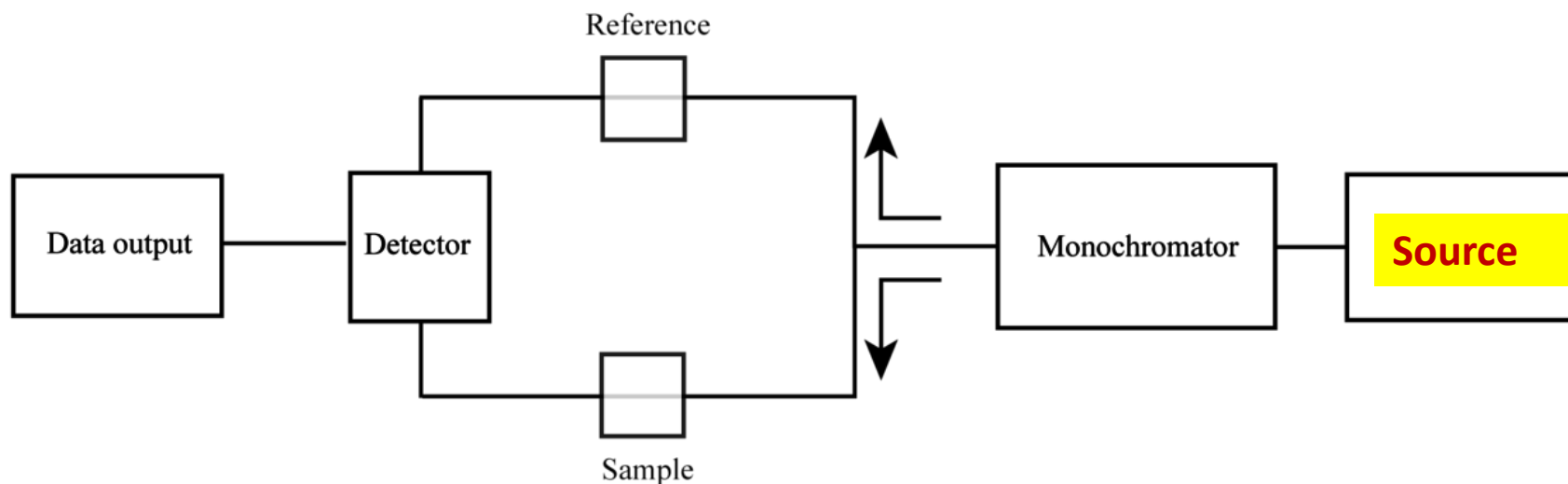
$$[A] = (\epsilon_{B2} A_1 - \epsilon_{B1} A_2) / ((\epsilon_{A1} \epsilon_{B2} - \epsilon_{A2} \epsilon_{B1})L)$$

$$[B] = (\epsilon_{A1} A_2 - \epsilon_{A2} A_1) / ((\epsilon_{A1} \epsilon_{B2} - \epsilon_{A2} \epsilon_{B1})L)$$

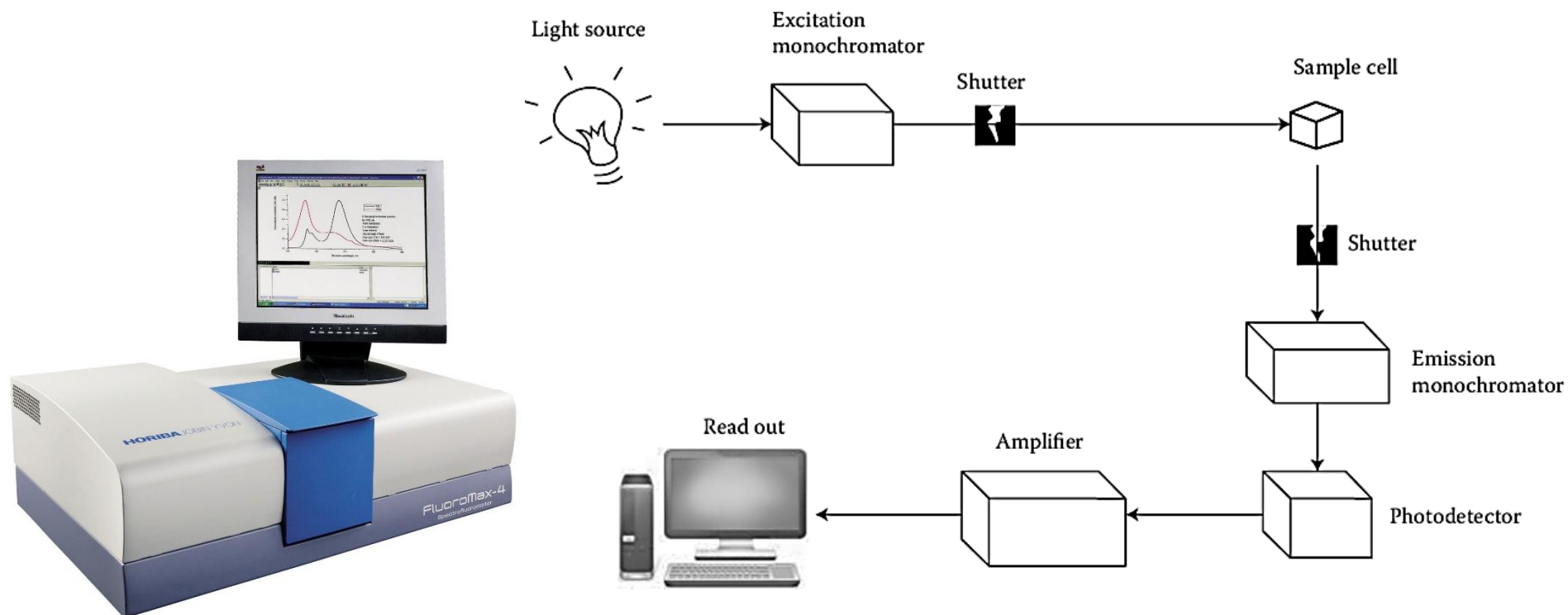
The isosbestic point indicates A in equilibrium with B.



Schematic diagram of a UV-vis spectrometer



Schematic diagram of a fluorometer



Vibrational spectroscopy (Infrared spectroscopy)

Vibrational spectroscopy



Diatomic molecule: A-B

$$V = \frac{1}{2} k x^2 = \frac{1}{2} k (R - R_e)^2$$

K: force constant with dimension
 N m^{-1}

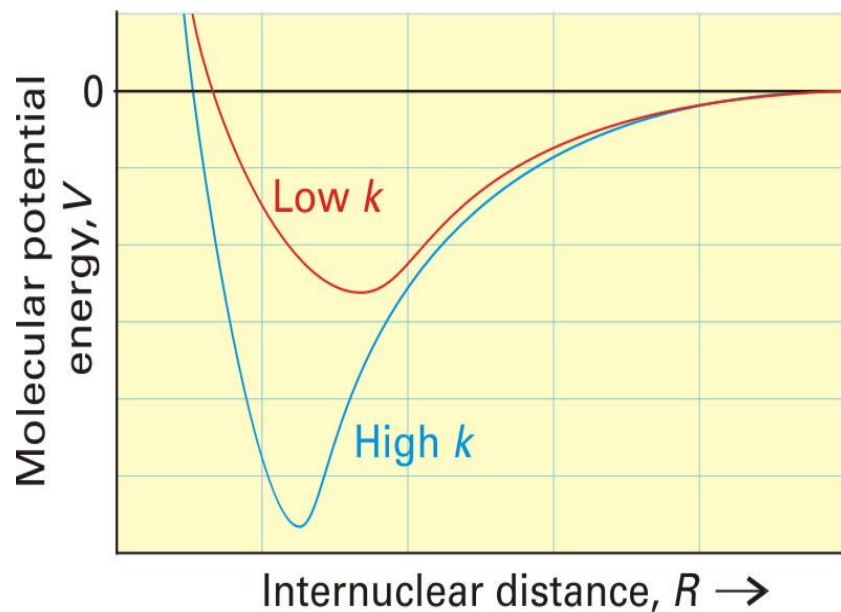
In the harmonic
approximation:

$$E_v = \left(v + \frac{1}{2} \right) h\nu; \quad v = 0, 1, 2, \dots$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}; \quad \mu = \frac{m_A m_B}{m_A + m_B}$$

In terms of wave number:

$$E_v = \left(v + \frac{1}{2} \right) hc\bar{\nu}, \quad v = 0, 1, 2, \dots$$



Vibrational spectroscopy



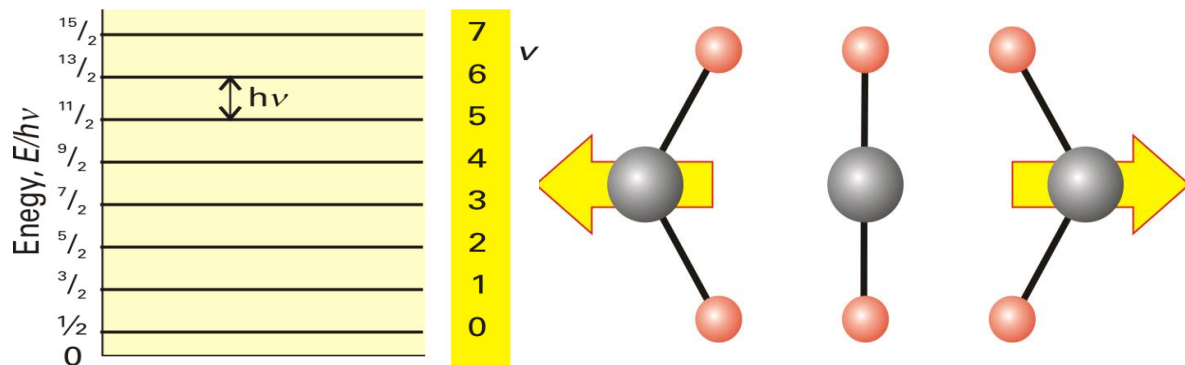
Gross Selection Rule: The electric dipole moment of the molecule must change during the vibration.

Homonuclear diatomics such as H_2 are infrared inactive.

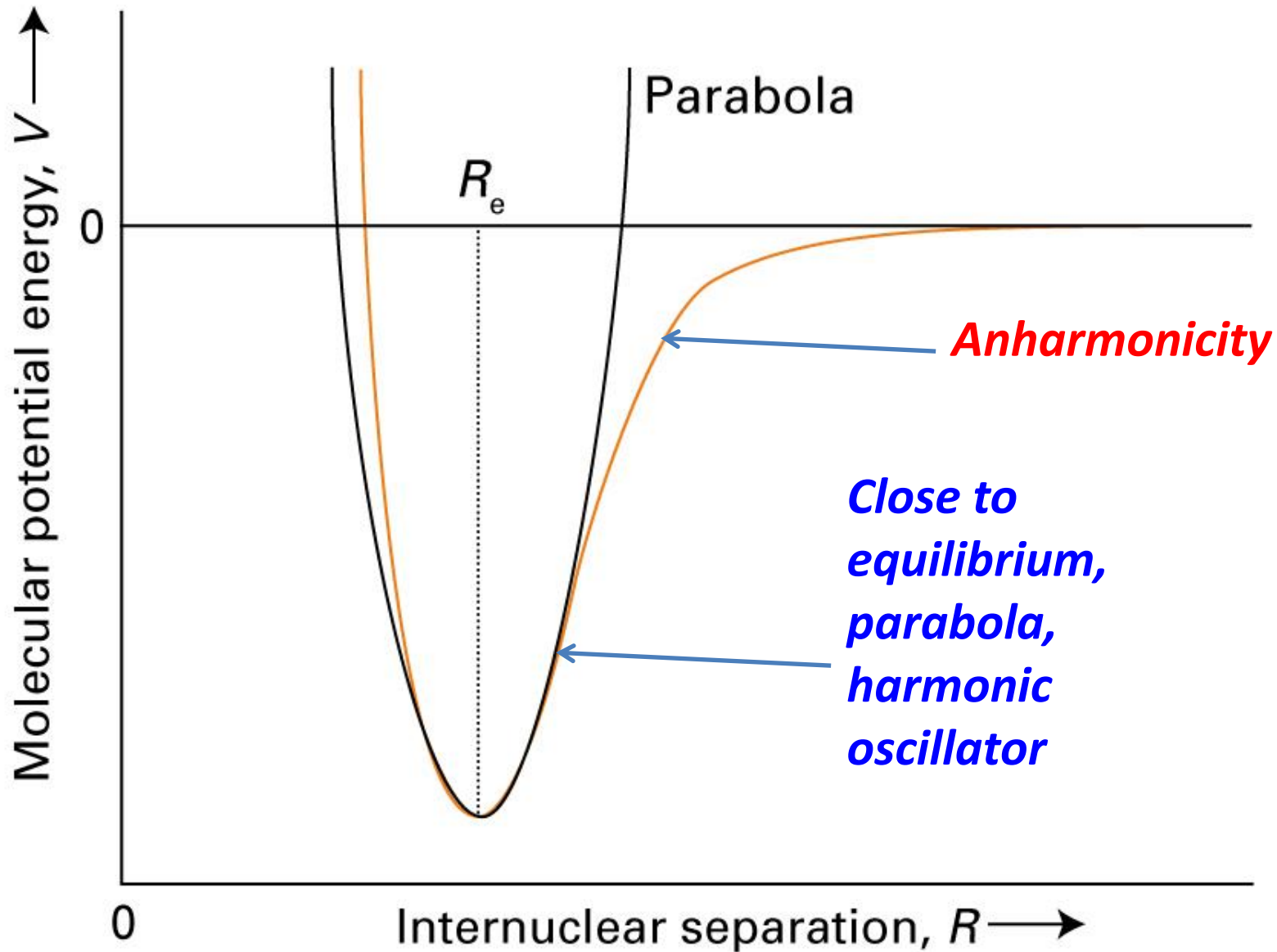
Specific selection rule: $\Delta v = \pm 1$

Most important transition from $v = 0$ to $v = 1$.

$$\Delta E = E_{v+1} - E_v = hc \tilde{\nu}$$



Vibrational spectroscopy



Anharmonicity:

$$E_v = (v + \frac{1}{2})hc\tilde{\nu} - (v + \frac{1}{2})^2 hc\tilde{\nu} x_e + \dots$$

x_e is anharmonicity constant.

*Additional weak absorption lines or **overtones** corresponding to $\Delta v = 2, 3, \dots$ appear. (near IR)*